

TUNGOKOCHEN, Russian Federation

WMO: 306640

Lat: 53.533N

Lon: 115.617E

Elev: 811

StdP: 91.95

Time Zone: 9.00 (E09)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-43.2	-41.2	-45.4	0.0	-42.8	-43.7	0.1	-41.1	6.3	-18.7	5.7	-16.6	0.1	220	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.8	27.4	16.8	25.6	16.5	23.8	15.8	18.7	23.8	17.7	23.0	16.9	22.0	2.7	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.0	13.4	20.6	16.0	12.5	19.6	15.0	11.7	18.8	56.6	23.7	53.5	23.0	50.7	21.9	22.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.6	6.4	5.5	DB	-46.0	30.3	1.6	2.5	-47.1	32.1	-48.0	33.6	-48.9	35.0	-50.0	36.9	(4)
(5)			WB	-45.6	20.5	1.5	1.0	-46.7	21.2	-47.5	21.8	-48.3	22.4	-49.4	23.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-4.9	-28.6	-23.5	-13.7	-1.7	6.5	14.0	16.4	14.0	6.5	-3.4	-18.6	-28.0	(6)
(7)		DBStd	17.02	5.60	5.57	7.13	5.19	3.78	3.40	2.47	3.32	4.23	5.28	7.66	6.12	(7)
(8)		HDD10.0	5932	1198	939	733	353	121	7	1	8	121	416	857	1179	(8)
(9)		HDD18.3	8505	1456	1172	992	602	368	134	69	138	356	674	1107	1437	(9)
(10)		CDD10.0	484	0	0	0	1	12	127	198	130	16	0	0	0	(10)
(11)		CDD18.3	16	0	0	0	0	0	4	9	2	0	0	0	0	(11)
(12)		CDH23.3	463	0	0	0	1	4	163	221	67	8	0	0	0	(12)
(13)		CDH26.7	81	0	0	0	0	0	38	37	5	1	0	0	0	(13)
(14)	Wind	WSAvg	1.7	0.8	1.1	2.0	2.7	2.8	2.0	1.7	1.6	2.0	1.9	1.1	0.6	(14)
(15)	Precipitation	PrecAvg	397	4	4	5	16	35	66	93	84	54	14	10	5	(15)
(16)		PrecMax	491	15	14	16	41	77	129	170	157	118	44	37	17	(16)
(17)		PrecMin	234	1	0	0	2	1	10	30	50	8	0	0	1	(17)
(18)		PrecStd	62	3	3	4	11	19	31	34	24	29	11	7	4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-9.3	-2.1	8.6	18.3	24.2	30.6	29.7	27.3	24.3	14.3	4.0	-4.8	(19)
(20)			MCWB	-11.0	-4.7	2.5	8.6	13.1	18.6	17.7	16.9	13.9	7.3	0.5	-6.7	(20)
(21)		2%	DB	-12.8	-6.5	5.2	13.6	21.2	27.3	27.6	25.2	20.5	10.9	-0.9	-11.6	(21)
(22)			MCWB	-14.0	-8.6	0.5	5.6	11.4	16.2	17.4	16.6	12.4	4.9	-3.2	-12.6	(22)
(23)		5%	DB	-15.8	-9.2	2.1	10.5	18.9	25.0	25.8	23.2	17.5	7.8	-3.5	-15.5	(23)
(24)			MCWB	-16.7	-10.8	-1.7	4.0	9.9	15.6	17.0	16.0	11.0	2.9	-5.1	-16.1	(24)
(25)		10%	DB	-18.6	-12.4	-1.1	7.8	16.2	22.8	24.2	21.2	14.9	5.1	-6.4	-18.4	(25)
(26)			MCWB	-19.2	-13.5	-3.9	2.3	8.4	14.8	16.6	15.4	10.0	1.3	-7.7	-18.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-10.9	-5.0	2.9	9.0	13.8	19.7	20.1	19.2	15.3	7.5	0.2	-6.5	(27)
(28)			MCDWB	-9.4	-2.3	8.3	17.0	23.5	27.2	24.9	23.3	20.6	13.9	3.8	-4.6	(28)
(29)		2%	WB	-14.2	-8.5	0.4	6.1	11.9	17.6	19.0	18.0	13.8	5.0	-2.9	-12.5	(29)
(30)			MCDWB	-13.2	-6.6	5.0	13.0	20.2	24.2	24.1	22.5	18.1	10.3	-1.2	-11.5	(30)
(31)		5%	WB	-17.0	-10.8	-1.7	4.3	10.3	16.6	18.1	17.0	12.3	3.2	-5.3	-16.1	(31)
(32)			MCDWB	-16.2	-9.3	2.0	10.1	17.6	22.8	23.4	21.2	16.3	7.5	-3.5	-15.5	(32)
(33)		10%	WB	-19.6	-13.6	-3.7	2.5	8.8	15.6	17.3	16.1	10.4	1.4	-7.8	-18.8	(33)
(34)			MCDWB	-19.1	-12.5	-1.1	7.4	15.5	21.5	22.6	20.0	14.9	4.9	-6.6	-18.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	14.6	19.6	19.7	14.2	14.3	14.6	13.8	13.1	13.7	13.5	14.7	13.4	(35)
(36)			MCDBR	16.7	20.3	20.0	19.0	20.4	19.7	18.4	17.3	18.6	17.0	14.9	15.2	(36)
(37)			MCWBR	16.0	18.3	15.4	12.1	11.4	9.8	9.2	9.2	10.8	11.3	12.5	14.3	(37)
(38)		5% WB	MCDWB	16.8	20.1	19.2	18.4	18.5	16.4	14.8	13.4	15.7	16.3	14.5	15.1	(38)
(39)			MCWBR	15.9	18.1	15.0	12.0	10.7	8.7	8.2	7.6	9.6	11.0	12.2	14.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.241	0.271	0.304	0.394	0.506	0.477	0.424	0.386	0.334	0.292	0.263	0.243	(40)	
(41)		taud	2.194	2.164	2.163	2.073	1.839	2.031	2.249	2.342	2.439	2.443	2.273	2.149	(41)	
(42)		Ebn at Noon	742	828	873	827	746	778	819	833	842	809	705	648	(42)	
(43)		Edh at Noon	62	90	114	144	192	161	127	109	87	69	58	53	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.25	2.38	4.09	5.21	5.45	5.89	5.45	4.63	3.55	2.37	1.36	0.94	(44)	
(45)		RadStd	0.04	0.05	0.09	0.31	0.37	0.40	0.48	0.26	0.29	0.16	0.08	0.03	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page